

B. Tech. in COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

III Year II Semester Syllabus (KR23)
INTRODUCTION TO AWS (23CD601PC)

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Pre-requisites/ Co-requisites:

1. 23CC403PC - Operating Systems Course

Course Objectives: The course will help to

1. Learn the fundamentals of Cloud Computing
2. Get an overview of fundamentals of AWS
3. Learn about AWS Virtual machines.
4. Understand IAM user roles, policies
5. Learn storing object S3

Course Outcomes: After learning the concepts of the course, the student is able to

1. Understand the fundamentals of Cloud Computing
2. Understand the basics of AWS
3. Implement AWS Virtual machines.
4. Implement IAM user roles, policies
5. Understand storing object S3

UNIT - I

Cloud computing fundamentals: Cloud Computing, Cloud essential characteristics, Cloud Deployment models, Cloud service models- Infrastructure as Service, characteristics of IaaS, Suitability of IaaS, pros and cons of IaaS, IaaS providers, Platform as Service, characteristics of PaaS, Suitability of PaaS, pros and cons of PaaS, PaaS Providers, Software as Service, characteristics of SaaS, Suitability of SaaS, pros and cons of SaaS, SaaS Providers.

UNIT - II

Introduction to AWS: Need of AWS, Benefit of AWS, Exploring AWS services, Creating an AWS free tier account, Ways to access AWS: CLI, Console, and SDKs, Explore Management Console and Configure AWS CLI.

UNIT - III

AWS Services: Exploring virtual machines, Monitoring and debugging a virtual machine, Changing the size of a virtual machine, Allocating a public IP address, Adding an additional network interface to a virtual machine, Rapid elasticity using Amazon EBS, Amazon EFS, Amazon VPC – including subnets, route tables and security groups.

UNIT - IV

Securing the system: AWS Identity and Access Management (IAM), Comparison of IAM user and root user, why IAM user preferred over root user, securing your AWS account's IAM user and AWS account's root user, Defining permissions with an IAM identity policy, Users for authentication, and groups to organize users, Authenticating AWS resources with roles, Automating operational tasks with AWS Lambda, Executing your code with AWS Lambda.

UNIT - V

Data Storing in AWS: Amazon S3, Storage classes of Amazon S3- Standard, Glacier, Glacier Deep Archive, Infrequent Access, Backing up your data on S3, Archiving objects to optimize costs, Versioning in S3, Using S3 for static web hosting, Protecting data from unauthorized access, Optimizing performance.

Manage AWS Billing: Free Tier, Pay-Per-Use Billing, Create a budget alert to keep track of your AWS bill, Alternate cloud providers.

TEXTBOOKS:

1. Amazon web Services in Action, Third edition, Andreas witting, Michael witting, Manning Publications, 2023.
2. Cloud Computing, Theory and Practice 2nd Edition, Dan C Marinescu, MK Elsevier publisher, 2019.

REFERENCE BOOKS:

1. Learn AWS Serverless Computing: A beginner's guide to using AWS Lambda, Amazon API Gateway, and services from Amazon Web Services by Scott Patterson.
2. Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud– by Mark Wilkins, Kindle E Textbook, 1st edition, 2019.